Fostering Cellular Agriculture

A Transparent Multi-Project Milestone Participation for Financing Research-to-Industry Scale-Up

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Version	1.3 / 1 September 2026
Revision scope	Empirical-underwriting boundary, development-credit evidence, and adverse-outcome discipline
Model case	Ten synthetic project claims / 60-month horizon
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Packaging creates no cash. Tranching reallocates risk. Honest rejection is a model output.

Abstract

Cellular agriculture requires patient capital across research, pilot, demonstration and first-industrial stages, yet early cash is sparse, technical failures are correlated and the legal source of investor repayment is often unclear. We define a candidate *Multi-Project Milestone Participation*: a limited-recourse claim on dated project draws, explicitly assigned success cash, recoveries and continuing principal across a disclosed set of projects. Later funding is callable only after stated milestones. Dependence is represented by complete joint states and a bounded set of physical probability measures rather than by multiplying independent project defaults.

The financial contribution is a source-separated, principal-conserving model that compares an unsupported callable core with two mutually exclusive variants: a fully funded first-loss/priority stack and a failure-contingent partial-credit guarantee. For the funded variant, acquisition and primary-funding cash, contractual asset principal, issued principal and direct costs remain separate ledgers. Contractual writeoff L, continuing asset principal O, and horizon issued-principal cash shortfall Q are reported as distinct observables; pooled cash is not used to invent causal impairment or recovery attribution. In a ten-claim synthetic experiment with 100 of reference principal, the core has central expected NPV of 0.661828 at an assumed 8% investor discount rate but adverse expected NPV of -18.717674. Pooling reduces central principal-loss ES95 by 10.05%, while ES99 diversification benefit is zero. Neither the funded stack nor the 30% guarantee restores robust investor adequacy under the declared terms. An additive version-0.2 frontier tests all 25 declared q-by-A terms on the same pool and finds no feasible point under the supplied synthetic mandates. The result is therefore a finite-grid rejection, not a financing claim or proof about untested terms. A separate empirical-underwriting boundary shows why exact synthetic calculations cannot establish expected loss, price or investor attractiveness without a frozen claim population, censoring discipline, realized cash evidence and independent market observations.

Keywords: cellular agriculture; financial engineering; milestone finance; project participation; portfolio dependence; partial-credit guarantee; blended finance; expected shortfall.
JEL classification: G11, G23, G32, G38, Q14.

Contents

1 Introduction

The objective of this research is practical: make it possible for institutional, mission and public capital to finance credible cellular-agriculture work from research into repeatable production without concealing the risk that capital is actually taking. The ethical purpose is to help build an industrial alternative to systems that confine and slaughter animals. That purpose raises the standard of evidence; it does not relax it.

The financing problem is not solved by naming a fund, adding projects, or promising that scale will arrive. Investors need a valuable asset: a defined property right to identifiable cash, a complete account of when capital is at risk, a distribution of loss and return that respects shared shocks, and terms under which the asset can be accepted or rejected. Public and philanthropic institutions need the same visibility from the opposite side: what risk they absorb, when cash can be called, whether support is additional, and whether a private return is being created by productive cash or subsidy.

Biomedical research-backed obligation proposals established an important intellectual precedent by combining uncertain development programs, staged asset value and structured claims [4–6]. The present design borrows the discipline of pooling, cash waterfalls and explicit guarantees, but it does not assume that cellular-agriculture projects possess comparable licensing markets, development probabilities, or debt capacity. It begins one level lower: every project must first connect to a common financial interface, and every unit of investor return must trace to an enforceable source.

This paper makes five contributions:

1. It defines the economic asset before the legal wrapper: a limited-recourse participation in dated, source-budgeted project cash, recovery and continuing exposure.
2. It gives one core construction and two alternative risk-allocation variants without claiming that tranching or guarantees create underlying project value, and supplies an explicit asset-to-liability bridge for non-par claims.
3. It evaluates the constructions under explicit joint physical states and probability ambiguity, preserving adverse witnesses rather than reducing the result to one expected return.
4. It turns non-investability into a useful output by stating measurable rejection conditions and the evidence needed for a live issuance.
5. It separates verified mechanics from empirical underwriting and defines the controlled cohort and transaction evidence needed before synthetic risk can become an investable estimate.

Key finding: the current terms are defined but not shown financeable

The synthetic core is barely positive at the central probability mix, yet fails under an admissible adverse mix. Central ES95 diversification exists but disappears at ES99. A funded junior layer introduces material prefunding drag, and a 30% guarantee improves investor NPV only by adding external provider cash; it still leaves a large adverse gap and an unfunded provider budget requirement. The version-0.2 frontier also rejects every one of 25 declared same-pool (q, A) candidates under the supplied synthetic mandate. This is evidence that the standard can expose a financing gap, not evidence about any live company or untested term.

2 The financing problem

2.1 Why research-to-industry scale-up resists ordinary debt

Cellular-agriculture projects cross qualitatively different stages. Shared science and platform research can create broad public and private option value while producing no near-term debt-service cash. Pilot and demonstration assets may generate operational evidence but remain exposed to biology, contamination, media supply, equipment integration, product qualification and buyer acceptance. First-industrial facilities add construction, commissioning, throughput, yield, cost and market risks at larger capital scale. Published techno-economic work illustrates why scale, media, mass transfer and process assumptions remain economically decisive, while reviews show that materially different assumptions still drive modeled outcomes [8, 10, 27]. Food-safety hazards and regulatory controls create another distinct evidence path [7]. Repeat facilities can become more debt-like only after these risks are evidenced and contractual revenues exist.

An ordinary fixed-rate loan presumes a sufficiently stable repayment source. A conventional venture investment can absorb more uncertainty but may be unsuitable for institutions seeking dated cash and bounded exposure. A single project can be too concentrated for either. The engineering task is therefore not to disguise early uncertainty as credit. It is to create a transparent progression of claims whose cash rights, loss states and evidence requirements mature with the underlying work. Dynamic-leverage research in biomedicine supports this sequencing: equity-like risk absorbs early discovery uncertainty and debt is introduced only as risk becomes measurable [15].

2.2 The five separations that prevent false value

The proposed standard maintains five separations throughout:

1. **Investor cash versus contractual principal.** Cash paid is a dated flow; principal is a contractual balance. At-par draws may make them numerically equal, but they are not conceptually interchangeable.
2. **Project cash versus external support.** Commercial, licensing, royalty and recovery receipts are produced by project rights. Guarantee payments are transfers from another balance sheet.
3. **Resolved loss versus continuing exposure.** A delayed or unresolved claim is not cash and is not automatically a realized loss.
4. **Expected loss versus price, provision and capital.** A provider's expected claim cost is not, by itself, a market premium, accounting provision, fiscal subsidy estimate or regulatory capital number [2, 3, 26].
5. **Mission impact versus financial return.** Animal displacement and welfare benefit require their own causal evidence. They do not substitute for investor cash flow.

These are not merely disclosure preferences. They are conservation boundaries. If a proposed return cannot be located on one side of a real contract, it is not part of the asset.

3 Related financial engineering and design departure

3.1 Research-backed obligations and correlation

Fernandez, Stein and Lo propose pooling biomedical-development assets in a bankruptcy-remote vehicle financed by senior debt, junior debt and equity, with repayment materially dependent on sale or licensing of development rights [6]. Later work applies the idea to smaller orphan-disease portfolios [5]. The relevance here is not either simulated return level. It is the architectural idea that heterogeneous research risks can be represented as assets only when the vehicle controls transferable value and the portfolio, waterfall, triggers and liquidity are modeled together.

Portfolio size alone is not diversification. Work on financing low-probability "long shots" emphasizes that even modest common dependence can undermine apparently safe senior claims and that tranching cannot convert a negative-NPV pool into a positive-NPV pool [9]. Correlated-development analysis reaches the same caution through explicit common-factor models [12]. Cellular agriculture has obvious common factors: related cell platforms, shared media and growth-factor suppliers, similar scale-up physics, common equipment vendors, regulatory pathways, buyers and follow-on capital markets. The present model therefore starts from complete joint states. It never obtains a portfolio default distribution by multiplying marginal defaults as if independent.

3.2 Guarantees and realized blended-finance analogues

The biomedical megafund guarantee literature shows why mean expected claims are an incomplete measure of the guarantor's commitment: adverse cash calls can be large even when mean cost appears modest [4]. The Global Health Investment Fund provides a realized analogue of a mission-oriented pool using milestone and royalty instruments with partial downside protection from philanthropic/public guarantors. Its reported mechanism absorbed the first 20% of losses and shared subsequent losses, leaving private investors exposed [28]. This supports the legitimacy of transparent partial loss sharing; it does not validate cellular-agriculture probabilities, recoveries or investor demand.

Blended-finance discipline requires a development rationale, additionality, minimum concessionality, commercial sustainability and transparency about the support used to mobilize private capital [11]. Accordingly, any first-loss or guarantee value in this paper is shown separately from project economics.

3.3 A documented public-credit precedent in cellular agriculture

Meatable announced in September 2024 that it had been awarded a EUR 7.6 million Innovation Credit from the Netherlands Enterprise Agency to improve cultivated-meat productivity and reduce costs before commercialization [13]. This is issuer-reported evidence of a named public development-credit award in cellular agriculture; it is not evidence of the exact legal recipient, individualized decision, cash receipt, draw schedule, probability of technical success, repayment cash, loss, or private-market value.

The official 2024 programme opening set a 3% annual base rate and a 15% fixed premium for technical-development projects [14]. Generic programme documents describe project reporting and cash-flow forecasts, include a template for a first-ranking pledge over identified project assets, and state that the credit, accrued interest and fixed premium are repayable unless relief is granted [24]. Current programme guidance, which postdates the announced award, describes relief after technical failure or loss of market prospects as a case-specific request that the agency may accept in whole or in part, not an automatic payoff formula or an award-specific term [25].

The precedent matters because it shows a candidate public-credit architecture for development risk rather than treating early research as ordinary bank debt. It also exposes the data a private pool would still need: the individualized award decision, borrower, approved budget, advances, milestone tests, exact repayment schedule, pledged assets, other financing, amendments, remission decision and actual settlement. The public record therefore supports mechanism research while remaining ineligible as a price, expected-return observation or calibrated loss history.

The adverse company outcome is now public too. On 19 December 2025 Agronomics reported that Meatable's board and shareholders had resolved to dissolve Meatable B.V. and related group companies, terminate operating activities and conduct an orderly wind-down after the company was unable to obtain continued funding; Agronomics also reported writing its own Meatable investment from a carrying value of GBP 11.9 million to zero [1]. This is evidence of a company wind-down decision and an equity-investor write-off, not evidence that the Innovation Credit defaulted, was accelerated or remitted, or produced any creditor recovery. The legal dissolution date, RVO claim status, outstanding public-credit principal, security enforcement and liquidation proceeds remain unidentified.

3.4 Where this proposal differs

The design does not begin with leverage or a rated senior tranche. It begins with a standardized claim ledger. It does not count company valuations, generic intellectual property, unused payer capacity or refinancing proceeds as operating cash. It does not assume a liquid secondary market for failed or partially developed assets. It does not force one instrument across all stages. Finally, its central result is allowed to be rejection.

4 Definition of the underlying asset

4.1 The project financial interface

Each candidate project is normalized through the same interface [19]. For project i, the interface must identify the items in table 1 before inclusion.

The investor owns a pro-rata, limited-recourse right to the admitted project claims, specifically assigned success cash, identified recoveries and any separately documented support. The investor does not automatically own a facility, company equity, company value, speculative intellectual property, a public sponsor's unused balance sheet, or a manager's promise.

4.2 Admission is a falsifiable contract test

A project is admitted only if the claim is unique, cash-source budgets do not overlap, milestone evidence can control draws, principal identities reconcile, the common calendar is explicit, and all material factor exposures are declared. Admission verifies representability; it does not validate the project or imply financeability. Missing evidence remains Unknown, never zero.

For a live pool, claim data must be authenticated against executed contracts, controlled data-room records, verifier certificates and settlement instructions. At minimum, legal diligence

Table 1: Minimum project financial interface

Component	Required representation	Reason for investors
Capital	Dated commitment, draw, fee and milestone schedule	Determines cash paid, liquidity need and the value of stopping later draws
State	Observable milestone, delay, stop, conversion and resolution states	Prevents narrative progress from replacing contractual conditions
Principal	Creation, repayment, conversion, write-off and continuing balance	Separates exposure from cash and realized loss
Success cash	Named commercial, licensing, royalty, capacity, offtake or exit right, with source budget	Establishes where non-principal return can legally come from
Recovery	Priority, amount, timing and evidentiary status	Determines loss-given-resolution without assuming salvage
Shared risk	Biology, scale-up, supplier, regulatory and buyer factor exposure	Makes dependence and concentration explicit
Evidence	Contractual, observed, estimated, synthetic or unknown status	Prevents unknown values from entering as zero

must establish transferability, perfection, priority, limited recourse, insolvency treatment and enforceability in each relevant jurisdiction.

5 The candidate instrument

5.1 Core: Multi-Project Milestone Participation

The core is one disclosed claim across a fixed set of project claims. Investors make callable commitments. Each project can draw only at dated milestones. Failure or non-achievement can stop later funding. Actual project receipts and recoveries pass through pro rata after explicit pool cost. The claim is not promised a fixed coupon independent of project cash; return arises only from identified rights.

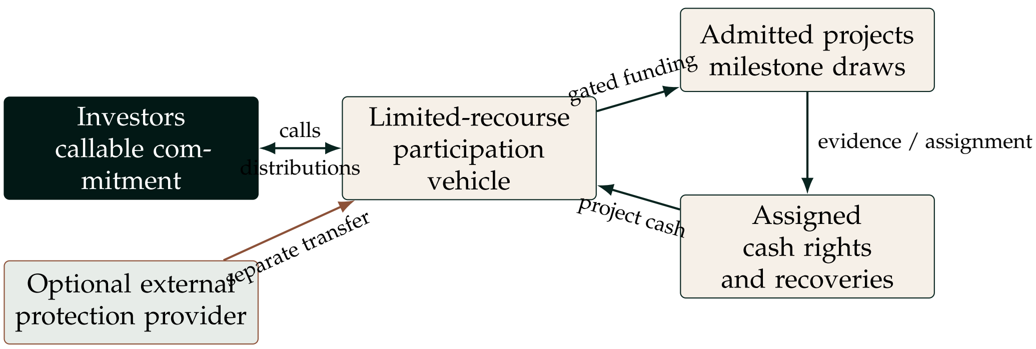

Figure 1: Economic architecture. Project cash and external support remain visibly separate.

5.2 Variant 1: fully funded first-loss and priority claims

Investors subscribe a reserve at inception. A funded first-loss claim occupies horizon principal-cash-shortfall coordinates beginning at zero; priority claims attach above it. Contractual-principal cash and unused reserve pay principal most senior first. Assigned non-principal cash pays priority up to a lifetime cap and then flows to the first-loss residual. Buyer direct costs and pool costs are separate pro-rata calls rather than tranche principal. Returned reserve is capital, not profit.

The asset-to-liability bridge does not assume that cash paid for an asset, contractual asset principal and issued principal are equal. Let A_{is} be project i's acquisition and primary-funding use in state s, excluding buyer direct cost. The funded limit is set project by project:

$$R_i = \max_{s \in \mathcal{S}} A_{is}, \qquad R = \sum_i R_i, \tag{1}$$

$$A_s = \sum_i A_{is}, \qquad U_s = R - A_s, \tag{2}$$

where R is both reserve subscription and issued tranche principal, and U_s is unused reserve returned at the horizon. This deliberately uses the sum of project maxima, not the maximum aggregate use in one state; projects can require liquidity in different states.

Let P_s be contractual-principal cash received by the analysis horizon. The liability ledger is

$$D_s = \min\bigl(R, P_s + U_s\bigr), \tag{3}$$

$$Q_s = R - D_s, \tag{4}$$

$$S_s = \max\bigl(0, P_s + U_s - R\bigr), \tag{5}$$

where D_s is principal cash distributable to issued claims, Q_s is their horizon principal-cash shortfall after the single aggregate canonicalization described below, and S_s is cash above unpaid issued principal that enters the declared non-principal waterfall. For a tranche with attachment a_j and detachment d_j,

$$Q_{js} = \min\bigl(\max(Q_s - a_j, 0), d_j - a_j\bigr). \tag{6}$$

Thus a 0–20 first-loss claim absorbs the first 20 of Q_s, while an attaching priority claim receives principal cash first.

Before tranche layering, the engine canonicalizes aggregate Q_s once: a non-negative residual no greater than $\tau(R) = 10^{-9} + 16\varepsilon_{\text{double}} \max(1, |R|)$ is set to zero. No tranche applies an independent tolerance. The resulting once-canonicalized Q_s is the liability observable used throughout the funded stack.

The asset ledger remains separate:

$$L_s = \sum_i \ell_{is}, \qquad O_s = \sum_i u_{is}. \tag{7}$$

Here L_s is contractual principal written off on resolved paths and O_s is contractual principal still outstanding on continuing paths. The model asserts no causal allocation, impairment identity or post-horizon recovery value linking L_s, O_s and Q_s. Pooled cash identifies the liability shortfall but not which asset economically caused it. Aggregate contractual-principal limit less A_s is reported only as a principal-limit capacity diagnostic: the limit need not equal principal created or acquired in that state, so the difference is not a price, discount, premium or fair value.

When contractual-principal cash and reserve return arrive in the same month, they have equal seniority. For a simultaneous surplus $S_{s,t}$, the engine records the source memo pro rata,

$$S^P_{s,t} = S_{s,t} \frac{P_{s,t}}{P_{s,t} + U_{s,t}}, \qquad S^U_{s,t} = S_{s,t} - S^P_{s,t}, \tag{8}$$

Table 2: Non-par hand checks for the exact asset-to-liability bridge

State	$R = A_s$	Principal limit	P_s	L_s	O_s	Q_s	Conclusion
Resolved 8-for-10	8	10	8	2	0	0	Asset loss; no liability cash shortfall
Performing 12-for-10	12	10	10	0	0	2	Liability cash shortfall; no asset loss
Continuing 8-for-10	8	10	3	0	7	5	Outstanding asset and cash shortfall remain distinct

All amounts are synthetic DEMO MILLION. These rows demonstrate accounting mechanics, not market price, impairment, recovery or investability.

and uses the same ratio for the paid-principal source memo. This is a deterministic convention for fungible cash, not observed provenance; both sources enter one waterfall, so the split changes neither total cash nor tranche entitlement.

This variant changes payment timing and priority. It does not change project receipts, contractual writeoff, continuing principal, common-factor dependence or diversification. Full prefunding can reduce call risk but introduces negative carry when capital waits in reserve. The ten-claim illustration below remains the documented at-par draw-equals-principal baseline; the explicit bridge above controls any non-par claim. No cross-version equivalence inference is made from that baseline.

5.3 Variant 2: failure-contingent partial-credit guarantee

The alternative guarantee attaches only to the untranched multi-project participation core defined in Section 5.1. Let g be the covered share, Γ the provider's legal cash cap and L_s final resolved loss. With full provider performance,

$$G_s = \min(gL_s, \Gamma), \tag{9}$$
$$L_s^{\text{retained}} = L_s - G_s. \tag{10}$$

The ten-claim experiment uses $g = 30\%$, $\Gamma = 30$, and month-60 settlement. Continuing exposure is not covered. The guarantee creates an external claim on a provider and a corresponding contingent liability for that provider; it does not create project revenue or diversification.

The funded stack and guarantee are intentionally alternative constructions. Combining them would require a new attachment, recovery, subrogation and no-double-recovery waterfall.

6 Formal model

6.1 Indexing, flows and balances

Projects are indexed by $i \in \{1, \ldots, n\}$, complete joint states by $s \in \mathcal{S}$, and months by $t \in \mathcal{T}$. Define:

- d_{ist}: investor-funded project draw;
- c_{ist}: explicit pool cost attributable to the state;
- r_{ist}^{P}: contractual-principal cash returned;

- x_{ist}: assigned non-principal success cash;
- ρ_{ist}: cash recovery after failure or resolution;
- u_{is}: continuing contractual principal at the horizon;
- ℓ_{is}: resolved contractual-principal loss;
- e_{is}: evidenced conversion or other non-cash extinguishment.

For state s, the unsupported investor net cash at month t is

$$CF_s(t) = -\sum_i d_{ist} - c_{st} + \sum_i \left(r^P_{ist} + x_{ist} + \rho_{ist}\right). \tag{11}$$

The cash-source identity is checked independently so that commercial, licensing/royalty, recovery and external-support sources equal the receipts assigned to the claim, without double counting principal classification.

For each claim and path, contractual principal must conserve:

$$b^{\text{open}}_{is} + b^{\text{created}}_{is} = r^P_{is} + e_{is} + u_{is} + \ell_{is}. \tag{12}$$

Continuing principal u_{is} is exposure: it is neither a horizon cash receipt nor a resolved loss. A payment in kind is not cash. Refinancing proceeds are liquidity unless they extinguish the investor's claim with real cash.

6.2 Investor return

At annual-effective hurdle h, state NPV is

$$V_s(h) = \sum_{t \in \mathcal{T}} \frac{CF_s(t)}{(1+h)^{t/12}}. \tag{13}$$

For probability vector p, expected NPV is $\mathbb{E}_p[V(h)] = \sum_s p_s V_s(h)$. The hurdle is an investor decision input, not a probability and not a model-produced fair discount rate. The experiment uses 8% for the core/common comparison, 15% for the funded first-loss class, and 8% for the funded priority class. These values are synthetic and not evidence of market demand.

6.3 Physical probability ambiguity

Instead of declaring one calibrated distribution, the model defines an admissible set

$$\mathcal{P} = \left\{ p : \underline{p}_s \le p_s \le \overline{p}_s, \sum_{s \in \mathcal{S}} p_s = 1 \right\}, \tag{14}$$

with optional event and marginal constraints in the extended engine. These are candidate *physical* probability measures. They are not risk-neutral probabilities, market-implied prices, posterior confidence intervals or regulatory scenarios.

For any metric linear in probabilities, robust endpoints are exact linear programs:

$$\underline{M} = \min_{p \in \mathcal{P}} \sum_s p_s M_s, \qquad \overline{M} = \max_{p \in \mathcal{P}} \sum_s p_s M_s. \tag{15}$$

Each endpoint can have a different witness p. Minimum and maximum rows in a table therefore cannot be assembled into a single composite scenario.

6.4 Loss tails and diversification

Let $L_s = \sum_i \ell_{is}$. For confidence level α, loss expected shortfall is the probability-weighted mean loss in the worst $1-\alpha$ tail, with fractional boundary-state allocation when required. The model also applies the same tail operator to NPV shortfall $\max(-V_s, 0)$.

The reported diversification benefit compares pooled loss ES with the sum of standalone-claim loss ES under the same declared probability measure:

$$D_\alpha = \sum_i \mathrm{ES}_\alpha(\ell_i) - \mathrm{ES}_\alpha(L). \tag{16}$$

A positive D_α is a measured distributional effect, not newly created cash. If $D_{0.99} = 0$, the severe common tail has defeated the measured pooling benefit at that level.

6.5 Provider economics and credit

The guarantee provider's research budget separates claim cash from the other ingredients of capacity:

$$\begin{aligned} &\underbrace{\mathrm{PV}(G - \text{subrogation})}_{\text{net claim cash}} + \text{administration} + \text{liquidity/capital cost} \\ &\qquad + \text{required provider return} = \text{declared funding requirement}. \end{aligned} \tag{17}$$

Investor premium capacity and provider requirement are computed separately. Their difference is the support gap. It is not a fair-value estimate or universal pricing equation. A credit-stress layer then distinguishes pledged collateral from unsecured provider exposure and models settlement default, including wrong-way states in which project losses and provider weakness co-occur.

7 Synthetic ten-claim experiment

7.1 Design and boundary

The experiment contains exactly ten claims, 100 of aggregate contractual reference principal, three possible milestone draws at months 0, 12 and 24, and a 60-month horizon [16]. Performing receipts total 164.65: 100 of principal plus 64.65 of assigned non-principal participation. Every amount, probability, recovery, factor loading, hurdle and provider term is invented. "DEMO MILLION" is a dimensionless monetary scale on a constant analysis-close basis; nominal cash is discounted only where an NPV or PV label is shown.

The fixture tests dated draw stops and cash accounting. It does not model actual milestone predicates, technical certificates, executed contracts or market prices.

7.2 Explicit common factors

The five factor notionals overlap and must not be added. Biology exposure totals 38, scale-up 64, supplier/media 60, regulatory/qualification 54, and buyer/acceptance 77. These figures describe reference principal touched by a factor, not loss and not separate assets.

Table 3: Synthetic pool composition by development stage

Stage	Claims	Principal	Share
Research	2	13	13%
Pilot	3	24	24%
Demonstration	2	18	18%
First industrial	2	29	29%
Repeat production	1	16	16%
Total	10	100	100%

Table 4: Shared-factor exposure by synthetic claim

Claim	Biology	Scale	Supplier	Regulatory	Buyer
Cell-line and media platform	✓		✓		
Serum-free media platform	✓		✓		
Cultivated beef pilot	✓			✓	✓
Cultivated poultry pilot	✓	✓			✓
Continuous-perfusion demonstration		✓	✓		
Food-grade scaffold pilot			✓	✓	✓
First-industrial beef facility		✓		✓	✓
First-industrial poultry facility		✓	✓		✓
Cultivated-fat demonstration	✓			✓	✓
Repeat modular production line		✓	✓	✓	✓

7.3 Joint states

Each state specifies all ten claim outcomes. *P* performs; *D* remains continuing with no horizon receipt; *E* stops after the first draw; *L* stops after two draws; and *C* is fully drawn and resolves with 10% recovery. Early-stop recoveries are stated fractions of cash actually drawn.

Table 5: Declared central joint-state measure

Joint state	Weight	Ten-claim path	Loss	Continuing
All perform	58%	`PPPPPPPPPP`	0.000	0
Biological/process shock	7%	`LDELPPPPLP`	13.130	7
Scale-up/commissioning shock	7%	`PPPLLPLLPL`	23.150	0
Supplier/media shock	6%	`DLPPDEPLPD`	10.675	32
Regulatory/qualification delay	5%	`PPDPPDDPDD`	0.000	54
Buyer/product-acceptance shock	7%	`PPDLPDLLLD`	17.970	31
Biology and scale-up compound	4%	`LDECLPLLLL`	35.560	7
Supplier/regulatory/buyer compound	4%	`DLDLDCCCLC`	58.005	24
Systemic commercialization freeze	2%	`CCCCCCCCCC`	90.000	0

Central pairwise realized-loss correlations range from 0.228251 to 0.938114. The admissible probability envelope moves state weights within declared bounds while preserving total probability.

8 Results

8.1 Unsupported callable core

Table 6 reports separately optimized probability-envelope endpoints and the declared central measure. A positive central expected NPV is not robust adequacy.

Table 6: Unsupported core results (DEMO MILLION, except probabilities)

Metric	Minimum	Central	Maximum
Investor cash contributed, incl. cost	88.303000	94.624000	99.503500
Principal cash receipts	57.989200	75.123400	88.386050
Non-principal success cash	34.130000	46.525500	55.633000
Total investor receipts	92.119200	121.648900	143.700550
Continuing principal exposure	1.150000	8.520000	18.740000
Resolved principal loss	2.152500	9.980600	21.525350
NPV at assumed 8% rate	-18.717674	0.661828	15.440326
Principal-loss ES95	19.074000	70.803000	90.000000
Principal-loss ES99	23.150000	90.000000	90.000000
Peak cumulative net outlay	88.147600	94.533000	99.487400

Resolved contractual-principal-loss probability is 16% / 37% / 60%. Negative-NPV probability is 28% / 42% / 60%. Each endpoint column except "central" is assembled metric by metric and may use a different admissible probability witness.

At the central measure, expected project cash remains source-separated: 77.1490 commercial, 40.8265 licensing/royalty and 3.6734 recovery. Pool cost is 1.0. There is no project-level explicit-support cash. The minimum-NPV witness assigns only 40% to all-perform and more mass to compound, buyer, regulatory and systemic states; it produces −18.717674. Full declared success participation cannot reach robust break-even.

8.2 Pooling changes the tail, not total cash

At ES95, the sum of standalone claim tails is 78.717 versus 70.803 for the pool: a reduction of 7.914, or 10.053737%. At ES99 both values are 90. The result is economically important in both directions. Moderate-tail pooling has measurable value, but the severe systemic state still reaches every claim and eliminates the measured diversification benefit.

Investor reading of the core

Central expected NPV of **+0.661828** is thin relative to model uncertainty. The admissible adverse expected NPV is **–18.717674**, negative-NPV probability is 42% centrally, and continuing exposure is 8.52. An investor is not being asked to accept a hidden industry bet: these are explicit reasons to reject the present terms until cash rights, evidence, price or external risk-bearing capacity change.

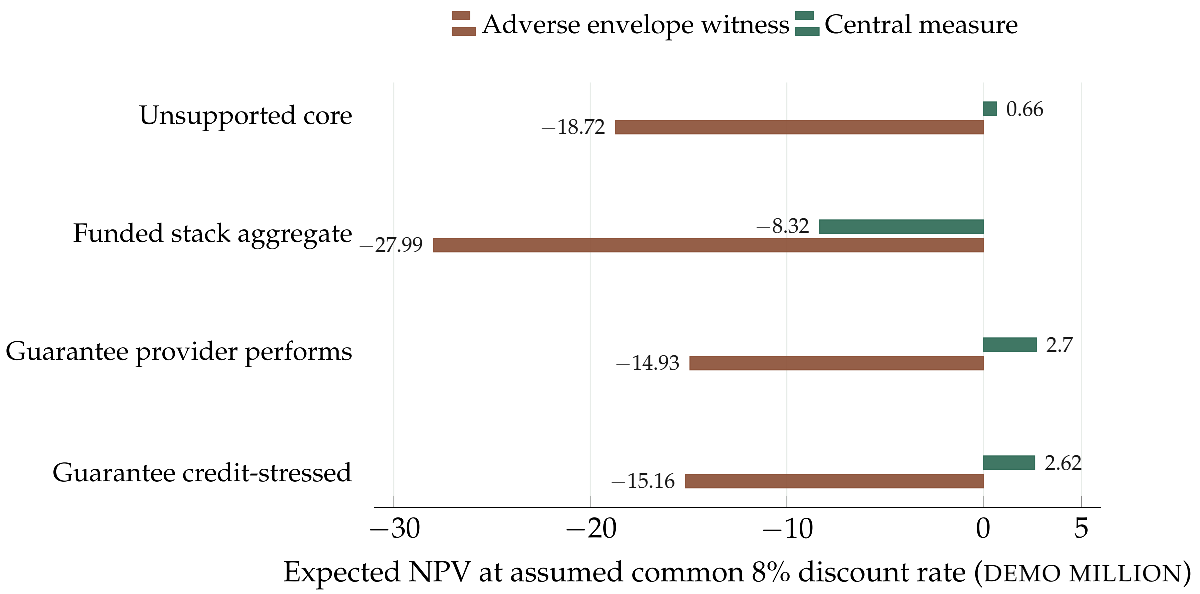

Figure 2: Expected NPV comparison. The funded stack aggregate uses an assumed common 8% discount rate; its individual classes have separate assumed rates. Guarantee values are before investor premium.

Table 7: What each construction changes

Construction	**Central NPV**	**Adverse NPV**	**Financial interpretation**
Unsupported callable core	0.661828	-18.717674	Milestone stops reduce unused draws; investor bears gross project loss and delay.
Funded stack, aggregate at 8%	-8.321750	-27.985093	Principal-cash priority and horizon-shortfall subordination become explicit, but full prefunding adds drag and creates no project value.
30% guarantee, provider performs	2.699617	-14.925982	External provider cash reduces retained resolved loss; premium is excluded.
30% guarantee, credit-stressed	2.617773	-15.161094	Wrong-way provider default reduces promised delivery in adverse states.

8.3 Comparison of constructions

8.4 Funded first-loss and priority variant

Investors subscribe 100 into the reserve at month zero and contribute 1 as a separate pool-cost call; project uses are funded from the reserve and are not counted again as investor contributions. Expected prefunding drag at the pool's assumed 8% discount rate is 8.983578 centrally. At their own assumed rates, the 0–20 first-loss class has central NPV -0.854614 at 15%; the 20–100 priority class has central NPV -13.926874 at 8%. Priority has negative NPV in every declared state. Aggregate stack NPV at the assumed common 8% rate is $-27.985093/-8.321750/6.715820$ across adverse, central and favorable expected cases.

The result is not an argument against subordination. It shows that subordination must be funded and priced. Asset L_s and O_s remain unchanged, while tranche principal-cash shortfalls sum to Q_s. The model does not attribute asset loss or outstanding principal causally to liability layers; rearranging the order of distributions cannot repair insufficient aggregate value.

8.5 Partial-credit guarantee and provider capacity

With full provider performance, central expected nominal guarantee payment is 2.994180 at month 60, increasing investor receipts from 121.648900 to 124.643080. Gross resolved loss stays 9.980600; investor-retained loss falls to 6.986420; continuing exposure stays 8.520000. The NPV improvement from 0.661828 to 2.699617 is the discounted value of an external transfer, not an improvement in projects.

The provider payment can reach 6.457605 as an expected value over the probability envelope, while single-path payment can reach 27. The modeled all-in support gap is 28.467332. Incorporating wrong-way provider credit raises the gap to 28.702444 and reduces central/adverse investor NPV to $2.617773/-15.161094$. The guarantee therefore fails both sides of the proposed market: it does not make the investor robustly adequate, and the investor cannot fund the provider's declared requirement with a non-negative premium consistent with its own target.

9 Investor value, risk and exposure

9.1 What could make the asset valuable

The candidate can become attractive only through real economic changes, not labels. Five channels are legitimate:

1. **Stronger assigned cash rights.** Enforceable commercial, licensing, royalty, capacity or offtake payments can increase expected and adverse cash without relying on company valuation.
2. **Better milestone control.** Earlier, independently verified stopping rules can reduce expected draws and downside liquidity while preserving value-producing paths.
3. **True diversification.** Projects using materially different technical routes, suppliers, jurisdictions and buyers can reduce common-tail concentration if the data support it.
4. **A lower issue price or different participation.** Price and success-cash sharing can align expected NPV with the buyer's hurdle, subject to issuer viability and source budgets.

5. **Explicit risk-bearing support.** A funded first-loss investor can take a defined issued-principal cash-shortfall layer Q; a capable guarantor can cover a defined resolved asset-loss layer L. Each transfers only the risk specified by its contract, and its cost, authority, liquidity and return requirement must be transparently financed.

These channels can be compared because the standard holds accounting constant. If a proposed term improves NPV while receipts, price, timing and external transfers are unchanged, the improvement is suspect.

9.2 Capital Mobilization Frontier v0.2: same-pool rejection and the conditional next channel

Version 0.2 asks a narrower investor question than the construction comparison above: with the ten project paths, event-constrained physical probability set, claim identifiers, lifetime market priority cap $B = 24$, assumed junior discount rate of 15%, assumed investor discount rate of 8% and zero junior target NPV held fixed, does any declared pair of success-cash participation q and funded junior issued principal A satisfy every supplied mandate? The tested grids are

$$q \in \{0.5, 0.625, 0.75, 0.875, 1\}, \qquad A \in \{10, 20, 30, 40, 50\},$$

so the result covers exactly 25 candidates. It is enumeration, not interpolation or continuous optimization.

The accounting layers must not be collapsed. The portfolio reports an aggregate project-outlay limit of 100 and an aggregate contractual asset-principal limit of 100. Their equality is a property of this retained at-par synthetic fixture, not an accounting identity. The v0.2 stack separately funds a reserve and total issued principal of $K = 100$; buyer-direct cost remains outside K. In state s, contractual writeoff L_s and continuing contractual principal O_s remain asset-ledger facts. Issued-principal cash shortfall is the liability fact

$$Q_s = K - D_s,$$

where D_s is principal cash distributable to the issued claims by the horizon. For a boundary A, the junior claim covers $[0, A]$ in Q-shortfall coordinates and the market claim covers $[A, K]$, with market issued principal $M = K - A$. Thus A allocates liability cash shortfall; it neither causes nor identifies asset loss.

The mandate required non-negative robust market NPV margin; worst-case $E[Q]/M \leq 10\%$, Q-ES95/M $\leq 50\%$, Q-ES99/M $\leq 60\%$, and $\Pr[Q > 0] \leq 35\%$; market negative-NPV probability $\leq 35\%$; NPV-shortfall ES95/M $\leq 60\%$ and ES99/M $\leq 70\%$; WAL ≤ 10 years; $A \leq 50$; and junior NPV concession ≤ 50. Aggregate NPV was deliberately left undeclared. A missing mandate does not silently bind.

Table 8: Two disclosed points from the rejected 25-candidate v0.2 frontier

$(q, A; M)$	**Robust market NPV**	**Worst** $E[Q]/M$	**Q-ES95/99 over** M	**Max** $\Pr[Q > 0]$	**Max** $\Pr[NPV < 0]$
$(1, 20; 80)$	-25.733095	27.377875%	87.5% / 87.5%	60%	100%
$(1, 50; 50)$	-5.569641	12.161%	80% / 80%	28%	54%

No candidate passed every declared mandate; consequently the feasible set, nondominated feasible set and minimum tested feasible q are all empty. The larger junior layer at $(1, 50)$ lowers market shortfall incidence, but still breaches the expected-Q, both Q-tail and negative-NPV screens. This finding applies only to the declared finite grid, fixed terms, physical probability set and invented mandate thresholds. It neither proves that an untested term fails nor establishes how an investor would price or buy the claim.

The next conditional channel is therefore not another relabelling of project loss. Holding the same $q = 1$, $A = 20$ claim and its future cash paths fixed, a priority-cap sensitivity selects $B = 24$, after which the issue-price/support module compares an investor's robust gross price ceiling $P^*(h)$ with the issuer funding floor

$$P_{\min} = \max(0, M + F - G),$$

where $F = 0$ is issuer cost and $G = 20$ is an invented maximum non-repayable support sensitivity. With $M = 80$, the synthetic floor is 60. The supplied independent hurdle cases produce the following arithmetic windows:

The priority-cap record uses a *separate, relaxed issue-price sensitivity mandate*: expected $Q/M \leq 30\%$, Q-ES95/M and Q-ES99/M $\leq 90\%$, $\Pr[Q > 0] \leq 60\%$, and WAL ≤ 5 years. Those are not the strict frontier's 10%, 50%/60%, 35%, and ten-year limits. The strict 25-candidate frontier rejects this same $(q, A; M) = (1, 20; 80)$ point. A pass under the separate sensitivity limits is therefore not a frontier pass. Price and support can change NPV and the issue funding identity, but they cannot change Q, Q-ES95/99, $\Pr[Q > 0]$, or WAL and cannot cure that fixed-risk failure.

Table 9: Conditional same-pool issue-price/support windows

Assumed rate h	Robust ceiling P^*	Minimum support $G_{\min}$	Shortfall to $G = 20$	Overlap with issuer floor 60
0%	74.575200	5.424800	0	[60, 74.575200]
5%	60.955502	19.044498	0	[60, 60.955502]
8%	54.266905	25.733095	5.733095	none
10%	50.313857	29.686143	9.686143	none
15%	41.898625	38.101375	18.101375	none
20%	35.170807	44.829193	24.829193	none

Here $G_{\min} = \max(0, M + F - P^*)$ is the modeled no-rights support needed just to create overlap, and "shortfall" is $\max(0, G_{\min} - 20)$. These are conditional source-and-price identities, not market observations. The reference gross buyer price $P_{\text{ref}} = 80$ exceeds every investor ceiling in the table. That reference, all six assumed rates and the support capacity are synthetic; settled support is zero; no support funding, escrow, authority, budget or settlement is evidenced; and no rate is empirically calibrated. Accordingly, the module finds arithmetic overlap only at 0% and 5%, finds none at the paper's assumed 8% investor discount rate or higher, and finds no funded/escrowed-support-covered window at any assumed rate. The separate relaxed screen does not reverse the strict frontier rejection. The module does not estimate fair value, discover a discount curve, show investor demand, prove legal enforceability, authorize execution, establish financing additionality, or establish capital mobilization. A live next step requires independent buyer-price and hurdle evidence, verified issuer uses and costs, and a legally authorized, funded and settlement-ready support source.

9.3 Exposure is multi-dimensional

No single rating-like number is appropriate at the present stage. Investors should receive at least:

- dated contribution and receipt distributions;
- expected NPV and probability of negative NPV at independently supported hurdles;
- resolved asset-loss tails for guarantee analysis, v0.2 issued-principal cash-shortfall expectation, incidence and ES95/ES99 for the funded stack, and NPV-shortfall VaR/ES at disclosed levels;
- continuing exposure, duration or weighted-average life where defined;
- project, stage, source, factor, buyer and provider concentrations;
- simultaneous-draw and peak cumulative liquidity requirements;
- adverse probability witnesses and factor attribution;
- provider expected payment, tail payment, collateral, unsecured exposure and wrong-way credit sensitivity; and
- separate mission additionality and impact evidence.

Expected return without these exposures is not an institutional description of the asset. Likewise, a low expected loss does not imply a tolerable tail, liquid price or small capital requirement.

10 From verified mechanics to empirical underwriting

10.1 Exact calculation is conditional, not calibrated

The current engine is exact about the question it is given. For each declared joint state it conserves dated cash and principal, and for each admitted probability set it finds the binding adverse witnesses. That precision is conditional on the state space, cash paths and probability restrictions. It does not make those inputs true. An expected loss computed from invented state weights is a synthetic sensitivity; a tail bound computed over analyst-declared probability limits is model ambiguity, not a sampling-confidence statement.

This distinction is the main investor-safety boundary. A tranche can be mechanically senior and still be impossible to underwrite if the population, censoring, recovery and dependence evidence are unknown. More projects, more simulations or a finer attachment grid cannot repair that problem. The empirical unit must first be a complete, economically de-duplicated claim history with known inclusion rules, comparable terms and a terminal or explicitly censored outcome.

10.2 The controlled cohort bridge

The project therefore defines a five-file Partial-Credit Claim-Loss Cohort Binder over verified Claim Ledger packages. Its population frame must enumerate resolved, unresolved, immature, disputed, denied, zero-loss and excluded members; exclusions remain visible and must have been frozen without reference to outcome. Every non-excluded row binds one verified claim root and one economically distinct risk unit. Open outcomes remain unknown rather than becoming zero, and descriptive amount ranges retain the full denominator. The repository now implements the closed five-file parser, exact hash binding, confined Claim Ledger reload and compiled population-frame Evidence Gate; this is structural admission only. Empirical realized-cash authority remains false because classification, method validity, term comparability, exclusion timing and source-status/date truth are not yet authenticated.

Even a future evidence-admitted binder has deliberately narrow authority. It may establish observed shortfall, provider claim, provider cash, unpaid claim and final writeoff experience for that controlled cohort. It does not by itself estimate a transferable probability of default, loss-given-default, common-factor dependence, fair premium, rating, portfolio tail or investor return. Those require a separate transfer model, effective-sample-size disclosure, uncertainty method, challenger specification and out-of-sample or later-vintage validation.

10.3 What the controlled transaction dossiers prove

Table 10: Controlled public-financing dossiers assembled for mechanism research, not calibration

Record	Publicly supported mechanism	Why it cannot identify price or loss
Liberation Labs, 2024–2025 [20]	Private secured notes with promised interest and a conversion feature for a fermentation facility.	Issuance clustering, all-in buyer cash, conversion, security, priority, recovery and expected-cash reconstruction remain incomplete.
Solar Foods Factory 01, 2022–2025 [22]	Amortizing bank facility for a gas-fermentation protein facility, a different operating-risk class from cultivated-meat tissue production, with capitalized commission, floating-rate terms, named export-credit guarantors and audited aggregate repayments.	The complete lender price, claim-level settlement, guarantee cash, default recovery and comparable adjustment bridge remain unavailable.
Meatable Innovation Credit, 2024 [1, 21]	Issuer-reported public-credit award directed to cultivated-meat productivity and cost reduction; a later investor RNS reports a resolution to dissolve and wind down the operating group after continued funding was unavailable.	The adverse company and equity-investor outcome does not identify RVO claim default, outstanding principal, acceleration, remission, priority, recovery or settlement; generic programme terms are not award-specific cash.

All three records improve the financial vocabulary of the standard. None enters the hurdle-evidence engine or calibrates the ten-claim pool. The current empirical result remains: no transaction-consistent discount-rate set and no transferable project-loss distribution have been identified.

10.4 Acceptance rule for an institutional calibration study

A future calibrated study must freeze one homogeneous claim family and observation horizon before viewing outcomes; retain every eligible member; publish missingness, censoring and concentration; bind cash and classification to dated source evidence; and separate estimation from validation. Its central and admissible physical measures must be reproducible from the retained cohort. Market price or hurdle evidence must come from independent settled cash or executable quotes rather than from the target model. A holdout, later vintage or predeclared challenger must show where transition, loss, recovery and dependence estimates fail. Until those conditions hold, the correct output is “not identified,” not a polished point estimate.

11 Conditions for a live issue

11.1 Contractual minimum

A live instrument requires a named issuer or vehicle, fixed reference claims, eligibility and substitution rules, commitment and termination dates, milestone verifier, payment agent, bank accounts, cash waterfall, loss and recovery definition, servicer replacement, reporting, audit rights, conflicts policy, security and perfection, insolvency treatment, transfer restrictions and dispute mechanism. Any support provider requires independent authority, appropriation or funding basis, claims procedure, settlement timing, cap, collateral and subrogation rules.

The economic label may ultimately map to a security, loan, participation, fund interest, derivative, insurance arrangement, public contingent liability or another regulated product depending on facts and jurisdiction. The paper deliberately does not select that classification.

11.2 Evidence minimum

Before solicitation, synthetic inputs must be retired by evidence with explicit lineage:

1. executed project cash rights and source budgets;
2. authenticated draw conditions and milestone histories;
3. project-stage transition, stop, delay, recovery and timing records;
4. joint-factor evidence rather than unsupported independence;
5. binding buyer, licensing, royalty, capacity or offtake terms;
6. current investor price/hurdle evidence that is not inferred from the same model;
7. provider mandate, capacity, pricing, collateral and credit evidence; and
8. legal, tax, accounting, prudential and insolvency opinions appropriate to the participants.

Three controlled public transaction dossiers in the repository are useful for mechanism research and question formation but fail the current evidence protocol and do not calibrate this model. "Controlled dossier" does not mean that every cited web source was successfully byte-retained; each source manifest reports retention status explicitly. No live project is admitted by this paper.

11.3 Issuance decision rule

The instrument should proceed only if all of the following hold under a frozen data room and model release:

- cash and principal reconcile in every path;
- all cash rights are unique, enforceable and transfer-compatible;
- milestones can actually stop later draws;
- adverse expected NPV and tail exposure satisfy identified investors at an evidenced price and hurdle;
- draw liquidity and continuing exposure are funded or explicitly retained;

- any support is authorized, financeable and additional;
- the provider remains capable under joint project/provider stress;
- concessions are the minimum necessary and are separately disclosed; and
- the financing causes incremental qualified capacity or knowledge relative to a documented counterfactual.

12 Falsification conditions and limitations

The investment thesis is falsified if any material condition below persists:

1. Named cash rights are absent, unenforceable, overlapping or insufficient.
2. Milestones cannot verifiably stop later draws.
3. Material shared shocks are omitted or results depend on silent independence.
4. Full contractual success participation still fails robust NPV.
5. Simultaneous draws or continuing exposure create an unfunded liquidity need.
6. Provider requirements exceed investor premium capacity.
7. Guarantee authority, funding, collateral control or enforceability is absent.
8. Market-hurdle evidence is circular, stale or incomparable.
9. Transferability, legal perfection, limited recourse or insolvency remoteness fails.
10. Concessionary support is hidden or is not demonstrably additional.
11. The financing does not cause incremental qualified capacity or credible animal displacement.

The current model is deliberately incomplete. It does not estimate live probabilities, fair value, liquidity discounts, tax, accounting classification, regulatory capital, endogenous behavior, strategic default, servicer failure, construction cost escalation, foreign exchange, commodity basis, dynamic refinancing, secondary-market price, moral hazard or investor demand. Joint states are finite and manually specified. Recovery is stylized. The 60-month horizon can classify economically valuable but unresolved claims as continuing exposure. Tail metrics are sensitive to state design and probability bounds. These limitations prevent the paper from supporting an issuance; they do not prevent it from defining what evidence an issuance would require.

13 Implementation and reproducibility

The analytical core is written in C++ and uses strict text configurations, deterministic scenario evaluation, exact cash-source and principal reconciliations, bounded-probability optimization, resolved asset-loss tails, issued-principal-cash-shortfall tails, NPV-shortfall tails, tranche waterfalls, guarantee payment, provider price decomposition and provider-credit stress. The maintained research repository includes model specifications, fixtures, command-line reporters, adversarial tests and verification records [16, 17, 23].

For the paper's reported snapshot:

- model date: 30 August 2026;
- exact portfolio: ten synthetic claims and nine complete joint states;
- monetary convention: synthetic DEMO MILLION, constant analysis-close basis;
- discounting: annual-effective, monthly exponents;
- reconciliation tolerance: 10^{-9} for modeled floating identities;
- verification status: the complete warnings-as-errors C++20 Emscripten 6.0.5 Release build and Node-driven suite pass 73 of 73 declared tests, including the strict five-file cohort loader, population-frame Evidence Gate, v0.2 asset-to-liability bridge, direct Claim Ledger integration, frontier core/parser/CLI regressions, issue-price browser target and published-WASM byte-parity check;
- immutable release identity: tag `fca-whitepaper-v1.3.0`; the tagged release manifest records the paper-relevant source, fixture and released-PDF hashes;
- portable execution: the release evidence reported for version 1.3 is produced by the pinned Emscripten container and Node, including every command-line report wrapper;
- paper toolchain: `latexmk` inside the Debian-based `documents-latex:bookworm` container.

The public browser interface is a readable analytical view of frozen synthetic results, and the same C++ kernel now has a strict WebAssembly build path. Browser interactivity does not itself prove parity or calibration: any published parity claim must remain tied to named structured-output fixtures, input hashes and retained cross-runtime tests.

14 Open-standard and responsibility commitments

Naturalehia retains no protocol fee, carried interest, instrument royalty or share of financed-company receipts for publishing or adopting this open standard. Underlying project claims may still assign capped commercial, licensing, royalty, capacity, offtake or exit cash to investors; without such rights there may be no investable asset. The zero-rent commitment reduces one conflict but does not improve cash flows or prove investment value.

The governing charter requires underwriting before advocacy, full visibility of risk transfer, transparent subsidy, no stronger claim than evidence, sponsor accountability, safety and food-integrity protection, conservative animal-impact attribution and an auditable go/no-go decision [18]. If rigorous analysis rejects a transaction, the mission is served by not spending scarce capital on terms that conceal failure.

15 Conclusion

The proposed Multi-Project Milestone Participation is a financial standard before it is a product. It defines a real candidate asset as a limited-recourse claim on dated, source-budgeted project cash and recoveries; represents unresolved exposure separately from loss; encodes shared shocks directly; and lets funded subordination allocate actual cash and horizon shortfall Q_s, or guarantee support transfer defined asset loss, without confusing either operation with value creation.

The synthetic experiment gives the appropriate first conclusion. Pooling provides some ES95 benefit, but common risk dominates ES99. The central core NPV is slightly positive, yet the adverse permitted NPV is materially negative. Fully funded priority creates drag, and the guarantee adds useful external cash but still cannot satisfy both investor and provider constraints. The additive v0.2 frontier rejects every one of the 25 declared same-pool (q, A) candidates. Its downstream issue-price/support sensitivity shows arithmetic overlap only under invented 0% and 5% discount-rate cases with uncommitted support, not at the assumed 8% investor discount rate, and no overlap is funded or escrowed. None of the current constructions is shown financeable.

The next serious step is empirical, not cosmetic: acquire enforceable project cash rights, milestone and recovery data, joint-risk evidence, investor price/hurdle observations and provider capacity terms in the common interface. With those facts, the same machinery can test whether a revised instrument is valuable, what risks it distributes across projects and institutions, and how much explicit catalytic support is genuinely required. That is the path from moral intention to finance capable of helping cellular agriculture reach industry.

Declarations

Funding. No external funding is represented for this paper.
Author economics. Naturalehia retains no protocol fee, carried interest, instrument royalty or share of financed-company receipts from adoption of the published standard.
Conflicts. No ownership, employment, advisory, consulting or other financial relationship with a named issuer, investor, guarantor or project has been disclosed to or relied upon by this research; no named party is endorsed or admitted. This statement does not substitute for signed author and adviser disclosures before any live transaction or third-party publication.
Data and code. Source, configurations, documentation and tests for this paper are fixed by release tag `fca-whitepaper-v1.3.0`. The numerical case is synthetic.
Responsibility. The paper is research, not professional advice or an offer. Independent qualified review is required before any transaction.

References

[1] Agronomics Limited. Dissolution of meatable. Regulatory News Service announcement 2963M, December 2025. URL https://www.lse.co.uk/rns/ANIC/dissolution-of-meatable-iooughimcr9uqi0.html. Published 19 December 2025; distributor copy accessed 1 September 2026.

[2] Basel Committee on Banking Supervision. The basel framework. Bank for International Settlements, 2026. URL https://www.bis.org/baselframework/BaselFramework.pdf. Downloaded framework current in August 2026.

[3] European Union. Regulation (eu, euratom) 2024/2509 on the financial rules applicable to the general budget of the union. Official Journal of the European Union, 2024. URL https://eur-lex.europa.eu/eli/reg/2024/2509/oj/eng.

[4] David E. Fagnan, José-María Fernandez, Andrew W. Lo, and Roger M. Stein. Can financial engineering cure cancer? *American Economic Review*, 103(3):406–411, 2013. doi: 10.1257/aer.103.3.406.

[5] David E. Fagnan, Austin A. Gromatzky, Roger M. Stein, José-María Fernandez, and Andrew W. Lo. Financing drug discovery for orphan diseases. *Drug Discovery Today*, 19(5):533–538, 2014. doi: 10.1016/j.drudis.2013.11.009.

[6] José-María Fernandez, Roger M. Stein, and Andrew W. Lo. Commercializing biomedical research through securitization techniques. *Nature Biotechnology*, 30(10):964–975, 2012. doi: 10.1038/nbt.2374.

[7] Food and Agriculture Organization of the United Nations and World Health Organization. Food safety aspects of cell-based food. Technical report, FAO and WHO, Rome, 2023.

[8] Corbin M. Goodwin, William R. Aimutis, and Rohan A. Shirwaiker. A scoping review of cultivated meat techno-economic analyses to inform future research directions for scaled-up manufacturing. *Nature Food*, 5:901–910, 2024. doi: 10.1038/s43016-024-01061-3.

[9] John C. Hull, Andrew W. Lo, and Roger M. Stein. Funding long shots. *Journal of Investment Management*, 17(4):9–41, 2019. URL https://joim.com/article/funding-long-shots/.

[10] David Humbird. Scale-up economics for cultured meat. *Biotechnology and Bioengineering*, 118(8): 3239–3250, 2021. doi: 10.1002/bit.27848.

[11] International Finance Corporation. How blended finance works. Institutional guidance, 2026. URL https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/how-blended-finance-works. Accessed 31 August 2026.

[12] Andrew W. Lo and Kien Wei Siah. Financing correlated drug development projects. *Journal of Structured Finance*, 27(1):17–34, 2021. doi: 10.3905/jsf.2020.1.114.

[13] Meatable B.V. Meatable was awarded EUR 7.6 million RVO innovation credit. Company announcement, September 2024. URL https://www.meatable.com/releases-announcements/meatable-awarded-eur-7-6-million-rvo-innovation-credit/.

[14] Ministry of Economic Affairs and Climate Policy of the Netherlands. Opening of the innovation credit module for 2024. Staatscourant 2023, 35780, 2023. URL https://zoek.officielebekendmakingen.nl/stcrt-2023-35780.html.

[15] Vahid Montazerhodjat, John Frishkopf, and Andrew W. Lo. Financing drug discovery via dynamic leverage. *Drug Discovery Today*, 21(3):410–414, 2016. doi: 10.1016/j.drudis.2015.12.004.

[16] Naturalehia. Ten-claim instrument analysis. Synthetic comparison, version 1.0, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/docs/TEN_CLAIM_INSTRUMENT_ANALYSIS_V1.md.

[17] Naturalehia. Fostering cellular agriculture: The financial instrument in brief. Open research specification, version 1.0, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/docs/FINANCIAL_INSTRUMENT_IN_BRIEF.md.

[18] Naturalehia. Responsible finance charter. Open project charter, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/docs/RESPONSIBLE_FINANCE_CHARTER.md.

[19] Naturalehia. Project financial interface. Open research standard, version 0.1, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/docs/PROJECT_FINANCIAL_INTERFACE_V0_1.md.

[20] Naturalehia. Liberation labs facility financing: Controlled public transaction dossier. Public-source normalization dossier with retained-source manifest, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/reference_transactions/liberation-labs-facility-financing-2024-2025/PUBLIC_TRANSACTION_DOSSIER.md.

[21] Naturalehia. Meatable dutch innovation credit: Controlled public transaction dossier. Public-source normalization dossier with source-retention manifest, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/reference_transactions/meatable-dutch-innovation-credit-2024/PUBLIC_TRANSACTION_DOSSIER.md.

[22] Naturalehia. Solar foods factory 01 financing: Controlled public transaction dossier. Public-source normalization dossier with retained-source manifest, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/reference_transactions/solar-foods-factory-01-facility-2022-2025/PUBLIC_TRANSACTION_DOSSIER.md.

[23] Naturalehia. Multi-project milestone participation: Candidate term sheet. Open research specification, version 1.0, 2026. URL https://github.com/savethebeesandseeds/naturalehia/blob/fca-whitepaper-v1.3.0/projects/fostering-cellular-agriculture/docs/MULTI_PROJECT_MILESTONE_PARTICIPATION_TERM_SHEET_V1.md.

[24] Netherlands Enterprise Agency. Innovation credit: Obligations and conditions of grant. Standard grant and security documents, July 2024. URL https://www.rvo.nl/files/file/2024-07/Innovatiekrediet-Verplichtingen-en-voorwaarden-verlening-versie-juli-2024.pdf.

[25] Netherlands Enterprise Agency. Innovation credit: Following your application. Official programme guidance, 2026. URL https://english.rvo.nl/subsidies-financing/innovation-credit/following-your-application. Accessed 1 September 2026.

[26] Office of Management and Budget. Circular a–11, section 185: Federal credit. Executive Office of the President of the United States, 2025. URL https://www.whitehouse.gov/wp-content/uploads/2025/08/s185.pdf.

[27] Laura Pasitka et al. Empirical economic analysis shows cost-effective continuous manufacturing of cultivated chicken using animal-free medium. *Nature Food*, 5:693–702, 2024. doi: 10.1038/s43016-024-01022-w.

[28] World Bank Group. GHIF: An innovative financing structure tackles neglected diseases around the world. Technical Report 112812, International Finance Corporation, World Bank Group, 2017. URL https://documents1.worldbank.org/curated/en/959631487668386111/pdf/112812-WP-GHIF-PUBLIC.pdf.

A Notation

Table 11: Principal notation

Symbol	Definition
i, s, t	Project, complete joint state and month.
d_{ist}	Project draw funded by investors.
b_{ist}	Contractual-principal balance.
r^P_{ist}	Contractual-principal cash returned.
x_{ist}	Assigned non-principal success cash.
ρ_{ist}	Recovery cash.
ℓ_{is}	Resolved contractual-principal loss.
u_{is}	Continuing contractual principal at horizon.
e_{is}	Evidenced conversion or other non-cash extinguishment.
L_s	Aggregate resolved contractual-principal writeoff, $\sum_i \ell_{is}$.
O_s	Aggregate continuing contractual principal, $\sum_i u_{is}$.
A_{is}	Acquisition and primary-funding reserve use for project i in state s, excluding buyer direct cost.
R_i	Project reserve limit, $\max_s A_{is}$.
R	Aggregate reserve subscription and issued principal, $\sum_i R_i$.
A_s	Aggregate state acquisition and primary-funding use, $\sum_i A_{is}$.
U_s	Unused reserve returned at horizon, $R - A_s$.
P_s	Contractual-principal cash received by the analysis horizon.
D_s	Principal cash distributable to issued claims, $\min(R, P_s + U_s)$.
Q_s	Horizon issued-principal cash shortfall, $R - D_s$; not an accounting impairment or recovery estimate.
S_s	Cash above unpaid issued principal, $\max(0, P_s + U_s - R)$.
a_j, d_j	Attachment and detachment of tranche j in Q_s-shortfall coordinates.
K	Funded reserve and top issued-principal stack detachment in the v0.2 frontier; equal to R for its supplied stack template.
A	Funded junior issued-principal boundary in Q-shortfall coordinates; not causal attribution of asset loss.
M	Market issued principal, $K - A$, excluding additional buyer-direct and pool-cost calls.
$CF_s(t)$	Investor net cash at month t in state s.
$V_s(h)$	State NPV at annual-effective investor hurdle h.
$p \in \mathcal{P}$	Admissible physical probability measure.
q	Participation share in assigned success cash.
B	Priority lifetime non-principal cash cap.
g	Guarantee share of final resolved loss.
Γ	Guarantee provider's legal cash cap.
G_s	Provider payment in state s.
P_0	Issue price or investor premium, when applicable.
D_α	Pooling diversification benefit at tail level α.

B Complete synthetic claim schedule

Table 12: Ten synthetic claims

#	Claim	Stage	Limit	Draws m0/12/24	Performing receipt
1	Cell-line and media platform	Research	6	2.4 / 1.8 / 1.8	11.40 at m60
2	Serum-free media platform	Research	7	2.8 / 2.1 / 2.1	12.95 at m60
3	Cultivated beef pilot	Pilot	8	2.8 / 2.8 / 2.4	14.00 at m54
4	Cultivated poultry pilot	Pilot	9	3.15 / 3.15 / 2.7	15.75 at m54
5	Continuous-perfusion demonstration	Demo	10	3.0 / 3.0 / 4.0	16.50 at m48
6	Food-grade scaffold pilot	Pilot	7	2.45 / 2.45 / 2.1	11.90 at m54
7	First-industrial beef facility	First industrial	15	3.75 / 5.25 / 6.0	23.25 at m48
8	First-industrial poultry facility	First industrial	14	3.5 / 4.9 / 5.6	21.70 at m48
9	Cultivated-fat demonstration	Demo	8	2.4 / 2.4 / 3.2	13.20 at m48
10	Repeat modular production line	Repeat	16	3.2 / 4.8 / 8.0	24.00 at m42
	Total		**100**		**164.65**

C Accounting and waterfall controls

Every evaluated state is subject to the following controls before reporting:

1. in the callable core, investor contributions equal project outlays plus explicit pool costs and any separately stated premium;
2. in the funded stack, investor contributions equal reserve subscription plus separate buyer-direct-cost and pool-cost calls; project acquisition and primary-funding uses consume reserve and are not counted again;
3. project receipts equal commercial plus licensing/royalty plus recovery plus other named project-source cash;
4. provider payments are external and never included in project-source cash;
5. principal returned, extinguished, continuing and lost equals opening plus created principal;
6. asset L_s and O_s are preserved separately, issued principal satisfies $R = D_s + Q_s$, and tranche principal-cash shortfalls sum to Q_s;
7. tranche distributions do not exceed cash available under the waterfall;
8. guarantee retained loss plus delivered provider payment equals gross resolved loss when full coverage terms apply, subject to the cap;

9. probability weights remain non-negative, within bounds and sum to one; and
10. each robust endpoint retains its optimizing probability witness.

The consolidated model reports zero residual for the declared source, principal, tranche and protection reconciliations in the ten-claim fixture. A zero accounting residual proves internal conservation only; it does not validate an economic input.

D Live-issuance diligence checklist

Table 13: Institutional work required beyond the research model

Domain	**Required evidence or decision**	**Failure consequence**
Reference assets	Executed claim schedule, unique rights, eligibility, substitution, transfer and servicing	Asset pool is undefined or double counted
Technical milestones	Objective predicates, verifier authority, cure, dispute and stop-draw mechanics	Milestone control has no economic force
Cash sources	Named payer, amount, date, source budget, priority and enforceability	Expected return is not an owned cash right
Dependence	Common-factor taxonomy, project mappings, joint observations and expert challenge	Pooling benefit is overstated
Recovery	Security, collateral, asset control, costs, timing and historical support	Loss is understated
Investor market	Comparable expected cash, issue price, constraints and evidenced hurdle	Adequacy test is circular
Protection provider	Authority, appropriation/funding, cap, collateral, claims and credit	Support may fail when needed
Legal structure	Classification, perfection, bankruptcy remoteness, limited recourse and tax	Waterfall or transfer may be unenforceable
Public purpose	Additionality, minimum concessionality, beneficiary, monitoring and clawback	Subsidy is hidden or unjustified
Animal impact	Qualified output, substitution baseline, attribution and assurance	Welfare claim is not permitted

E Interpretive boundary

This white paper is the concise technical synthesis of a larger open research program. Detailed format specifications, test fixtures, verification reports and candidate transaction dossiers remain in the repository. They are intentionally not reproduced here. The paper controls the public financial argument; the machine-readable sources control the implemented mechanics. If the two conflict, the stricter evidence boundary and the reproducible source calculations govern until the paper is corrected.